ESP32-S3 SuperMini Development Board

Dual-Core Xtensa LX7 to 240 MHz · 2 MB PSRAM · Wi-Fi and Bluetooth 5 LE · Native USB-C · 35 GPIO · LiPo Charger

ESP32-S3FH4R2 | 35x GPIO | 240 MHz DUAL CORE | 512 KB SRAM | 4 MB FLASH | 2 MB PSRAM | 23.5 x 18 mm | 3.3 V LOGIC

1 FEATURES

- **ESP32-S3FH4R2**: **two** Xtensa LX7 32-bit cores to **240 MHz**, with 128-bit vector instructions for DSP and neural-network work
- **4 MB in-package quad-SPI flash and 2 MB in-package quad-SPI PSRAM**, plus 512 KB SRAM, 384 KB ROM and 16 KB RTC SRAM
- **Wi-Fi**: IEEE 802.11 b/g/n on 2.4 GHz, TX to 21 dBm. **Bluetooth 5 LE** with Mesh, Long Range coded PHY and 2 Mbps high speed
- **35 GPIO broken out**: 18 castellated pads, nine per rail, plus **20 numbered pads on the underside**
- **18 ADC channels** on GPIO1-GPIO18 across two 12-bit SAR converters, and **14 capacitive touch channels** on GPIO1-GPIO14
- **Native USB over USB-C**: USB Serial/JTAG for flashing and debug, or full-speed **USB OTG** so the board can be a keyboard, mouse, MIDI or mass-storage device
- Onboard **single-cell lithium charger** with **B+ / B- pads** underneath, a charge-status LED and a **BOOST** solder jumper that lifts charge current from 100 mA to 300 mA
- **WS2812 RGB LED plus a discrete LED, both on GPIO48.** There is no independent plain user LED
- 3x UART, 2x I2C, 4x SPI, 2x I2S, TWAI (CAN 2.0), SD/MMC host, RMT, MCPWM, pulse counter, and an LCD / DVP camera interface
- Hardware AES-128/256, SHA, RSA, ECC, HMAC, digital signature, RNG, secure boot and flash encryption
- **Deep-sleep 8 uA** at chip level with the RTC timer and RTC memory retained

2 APPLICATIONS

- Edge AI and TinyML inference
- Camera and LCD projects needing PSRAM
- ESPHome and Home Assistant nodes
- Bluetooth LE peripherals and gateways
- USB HID: keyboard, mouse, MIDI, gamepad
- I2S audio capture and playback
- Battery-powered Wi-Fi sensors and loggers
- Wearables and space-constrained builds

3 DEVICE INFORMATION

SUPPLIED AS	ORDER CODE	CONFIGURATION
	**UNSOLDERED** **ESP32-S3-MINI-1PK**	Board plus two loose 2.54 mm male header strips. Fit straight, right-angle or low-profile headers, direct-wire the pads, or reflow the board onto a carrier by its castellated edges.

Board sourcing. Chip identified from the package marking as **ESP32-S3 FH4R2**, so 4 MB in-package flash **and 2 MB in-package PSRAM** in the QFN56 package. Electrical data in this document describes the chip silicon, not the assembled board. Every unit is inspected and powered up before dispatch, but that is a basic functional check only. Prototyping and hobby use - read Section 5.2 before designing this board into anything.

Most of the IO is on the underside. Only 18 pads reach the castellated edges. The other 20 GPIO are numbered solder pads on the back of the board, so they are unreachable once it is seated flat on a breadboard. Plan for a carrier, a socket or fine wire if you need them. See Section 4.

3.3 V logic - not 5 V tolerant. Applying 5 V to any GPIO will damage the chip. The **5V pad** is the only input rated above 3.3 V; The manufacturer rates it 3.3-6 V. **Do not power the 5V pad while USB-C is plugged in** - the manufacturer is explicit that only one source may be used at a time. **GPIO45 and GPIO46 are strapping pins sitting on the underside**, exactly where a carrier board is most likely to touch them: see Section 6.

4 PIN CONFIGURATION AND FUNCTIONS



TOP VIEW
USB-C at top · 18 castellated pads

BOTTOM VIEW
same way up · 20 numbered underside pads

Power | Ground | Digital I/O | Analog capable | Strapping pin - see Section 6

18 castellated pads, nine per rail, 2.54 mm pitch, plus **20 numbered pads on the underside**, the **B+ / B-** battery pads and the **BOOST** charge jumper. Both views are drawn the same way up, so each row lines up across the two. Pin functions are tabulated in Section 4.1.

4.1 PIN FUNCTIONS

PIN	TYPE	CASTELLATED EDGE PADS - DESCRIPTION
5V	PWR	USB VBUS or external supply into the onboard 3.3 V regulator and the charger. **Do not drive this pad while USB-C is connected to another source.** The manufacturer states 3.3-6 V is acceptable; Section 5.3 gives the range NZN recommends.
GND	GND	Ground return. One pad only; bond to the supply ground by the shortest practical path.
3V3	PWR	Output of the onboard 3.3 V regulator. May also be driven as a regulated 3.3 V input when 5V, B+ and USB are all unpowered.
GPIO1-2	I/O	ADC1_CH0 and ADC1_CH1, TOUCH1 and TOUCH2, both in the RTC domain so both can wake the chip from deep sleep. No reset-time commitment: the two cleanest analog inputs on the board.
GPIO3	I/O	**Strapping pin.** Sampled at reset to choose the JTAG source (built-in USB Serial/JTAG or the external MTxx pins). Floating at reset and ignored unless the matching eFuse is burned, but do not fit a hard pull here. Also ADC1_CH2, TOUCH3 and RTC.
GPIO4-7	I/O	ADC1_CH3 to ADC1_CH6, TOUCH4 to TOUCH7, all RTC capable. GPIO4-GPIO7 are also the IO MUX defaults for FSPI (FSPIHD, FSPIWP, FSPICLK, FSPID) if you use the general-purpose SPI peripheral at full speed.
GPIO8-9	I/O	ADC1_CH7 and ADC1_CH8, TOUCH8 and TOUCH9, RTC capable. The Arduino core defaults **I2C SDA to GPIO8 and SCL to GPIO9** on an S3, so most I2C examples land here unmodified.
GPIO10	I/O	ADC1_CH9, TOUCH10, FSPICS0, RTC capable. Last channel on ADC1.
GPIO11-13	I/O	ADC2_CH0 to ADC2_CH2, TOUCH11 to TOUCH13, RTC capable. **ADC2 cannot be read while Wi-Fi or Bluetooth is running**, so treat these as digital pins in any connected design.
GPIO43	I/O	Marked **TX**. U0TXD, hardware UART0 transmit, and the pad the ROM boot log comes out of. Free for general use once the console runs over native USB.
GPIO44	I/O	Marked **RX**. U0RXD, hardware UART0 receive. Outside the RTC domain, so it cannot wake the chip.

PIN	TYPE	UNDERSIDE PADS - DESCRIPTION
GPIO14	I/O	ADC2_CH3, TOUCH14, RTC capable. Last of the fourteen touch channels.
GPIO15-16	I/O	ADC2_CH4 and ADC2_CH5, U0RTS and U0CTS, RTC capable. Also **XTAL_32K_P and XTAL_32K_N**: using them as IO rules out an external 32.768 kHz crystal.
GPIO17-18	I/O	ADC2_CH6 and ADC2_CH7, IO MUX defaults for UART1. **GPIO18 is the last RTC-domain pad**; everything numbered above it is dead to deep-sleep wake.
GPIO21	I/O	General-purpose digital IO, RTC capable, no analog and no reset-time role. One of the cleanest spare pins on the board.
GPIO33-37	I/O	Free on this part. On S3 modules with **octal** flash or PSRAM these five carry SPIIO4-SPIIO7 and SPIDQS and cannot be used at all. This board's FH4R2 uses quad memory, so they are ordinary digital IO - which is why the manufacturer brings them out.
GPIO38	I/O	General-purpose digital IO. Outside the RTC domain: no deep-sleep wake, no analog, no touch.
GPIO39-42	I/O	JTAG **MTCK, MTDO, MTDI, MTMS** in that order. Usable as ordinary digital IO, and normally are, because the S3 debugs over its built-in USB Serial/JTAG instead. Reserve them if you intend to attach an external JTAG probe.
GPIO45	I/O	**Strapping pin.** Selects the VDD_SPI rail that powers the in-package flash and PSRAM. Get this wrong at reset and the board does not boot at all. See Section 6 before wiring anything to it.
GPIO46	I/O	**Strapping pin.** Controls ROM message printing at boot. Internal pull-down at reset. Usable afterwards, but nothing attached to it may hold it high through reset.
GPIO47	I/O	General-purpose digital IO. No RTC, no analog, no touch, no reset-time role.
GPIO48	I/O	Drives the **WS2812 RGB LED** and the discrete LED beside the USB connector. Both are on the same pad, so they cannot be used independently. Free as an ordinary pin if you ignore the LEDs.
B+ / B-	PWR	Terminals of the onboard single-cell lithium charger, two large round pads under the USB-C connector. B- is common with GND. The cell charges whenever USB-C is connected and powers the board when it is not.
BOOST	CFG	Solder jumper beside the battery pads. Bridged, it lifts charge current from **100 mA to 300 mA**. The manufacturer restricts that to cells of **500 mAh or larger**.

Not brought out. GPIO0 is under the BOOT button and has no pad. **GPIO19 and GPIO20** are the native USB D- and D+ lines and are committed to the USB-C connector, so unlike some S3 boards you cannot reclaim them and accidentally kill the console. **GPIO26-GPIO32** carry the in-package SPI flash and PSRAM inside the QFN56 package. Everything else the package exposes is on this board: **35 GPIO**, being GPIO1-GPIO18, GPIO21, GPIO33-GPIO48.

Peripheral mapping. The ESP32-S3 GPIO matrix routes UART, I2C, I2S, SPI, RMT, LED PWM, MCPWM, TWAI, pulse-counter, camera and LCD signals to almost any exposed GPIO. The names above are the IO MUX defaults from the chip pin table, not fixed assignments. Only the ADC, touch and RTC functions are tied to specific pads.

Read the underside, not the top face. The back of the board silkscreens the GPIO number of every pad, including all eighteen castellated ones, which makes it the fastest and least ambiguous way to read this board. The top face carries only **48** beside the two LEDs and **BAT** beside the charge LED - both annotate indicators, not pads. There is also one unmarked plated hole beside GPIO8 that the manufacturer does not document; treat it as a test point, not an IO.

4.2 FIRST USE

1. **Connect a USB-C data cable.** The board enumerates on its own. No driver, no bridge chip. A charge-only cable powers the board but never appears as a port.
2. **Arduino IDE: select the board first.** Install the official **esp32** board package, then Tools > Board > **ESP32S3 Dev Module**. **USB CDC On Boot, USB Mode, PSRAM and Flash Size do not exist in the Tools menu until an S3 board is selected** - going looking for them first is a dead end. Take care not to pick plain "ESP32 Dev Module", which sits beside it in the list and has no native USB.
3. **Set USB CDC On Boot to Enabled** or the Serial Monitor stays blank, and **PSRAM to QSPI PSRAM** or the 2 MB is never mapped. Both default to off.
4. **Select the port and upload.** The **first** connection may want the BOOT hold before a port appears, which is normal on a new board: hold **BOOT**, tap **RST**, release RST, then release BOOT. Full recovery steps are in Section 7.4.
5. **Check PSRAM actually came up.** `ESP.getPsramSize()` should report about 2 MB. Zero means the board menu setting did not take.

```
// Cycle the RGB LED and confirm PSRAM is mapped.
// Library: Adafruit NeoPixel. Board: ESP32S3 Dev Module
// Tools > PSRAM: QSPI PSRAM, USB CDC On Boot: Enabled
#include <Adafruit_NeoPixel.h>

#define RGB_PIN 48       // WS2812 and the discrete LED
Adafruit_NeoPixel px(1, RGB_PIN, NEO_GRB + NEO_KHZ800);

void setup() {
  Serial.begin(115200);  // over USB-C
  px.begin();
  px.setBrightness(40);  // full brightness is harsh
  delay(1500);
  Serial.printf("PSRAM %u bytes, heap %u bytes\n",
                ESP.getPsramSize(), ESP.getFreeHeap());
}

void loop() {
  px.setPixelColor(0, px.Color(255, 0, 0)); px.show(); delay(400);
  px.setPixelColor(0, px.Color(0, 255, 0)); px.show(); delay(400);
  px.setPixelColor(0, px.Color(0, 0, 255)); px.show(); delay(400);
}
```

5 SPECIFICATIONS

5.1 ABSOLUTE MAXIMUM RATINGS

PARAMETER	MIN	MAX	UNIT
Supply voltage on chip power pins [1]	-0.3	3.6	V
Voltage on any GPIO pad	-0.3	3.6	V
Cumulative IO output current	-	1100	mA
Storage temperature, T_{STORE} [2]	-40	105	°C

Stresses beyond these ratings may cause permanent damage. These are stress ratings only; functional operation at these conditions is not implied. [1] ESP32-S3FH4R2 silicon ratings, not board ratings. The **5V** pad feeds the onboard regulator and charger; The manufacturer rates that input at 3.3-6 V and publishes nothing further about it. [2] The in-package PSRAM sets this limit. Parts without in-package memory are rated to 150 °C; this one is not.

5.2 SOURCING, HANDLING AND LIMITS OF THIS DATA

Board sourcing. NZN Electronics buys this board through established supply channels rather than the cheapest listing available. It is the manufacturer's build of the widely copied ESP32-S3 SuperMini layout. The manufacturer does not publish lot traceability, qualification reports or change notification, and component choices such as the 3.3 V regulator and the charge IC vary between production batches.

Check any manufacturer figure before you rely on it. the manufacturer's own description of this board calls the ESP32-S3 a "32-bit RISC-V single core processor with a running frequency of up to 160 MHz", lists 400 KB of SRAM, quotes 43 uA deep sleep and counts "11xGPIO (PWM), 4xADC". Those are ESP32-C3 figures and none of them describe this part: it is a **dual-core Xtensa LX7 to 240 MHz with 512 KB SRAM**, 8 uA deep sleep and 35 GPIO. the manufacturer's resource line also omits PSRAM entirely - it does not deny it, but the only thing on that page telling you **2 MB of PSRAM is present** is the chip model, which the manufacturer does give correctly as **ESP32S3FH4R2**. Two lines on that page are right and worth keeping: the 23.50 x 18 mm size, and "Onboard RGB, Blue LED: Common GPIO 48 pin". Every figure in this document comes from the ESP32-S3 series datasheet and from the board itself, not from the supplier's listing.

What NZN checks, and what the numbers describe. Every board is visually inspected and powered up to confirm it enumerates over USB and runs. That is a basic functional check, not a qualification test: it does not verify tolerance, temperature range, RF performance, charger behaviour, regulator capability or service life. Every electrical figure here describes the **silicon**, not the assembled board, and is not guaranteed by NZN or by the manufacturer. Board-level behaviour is bounded by the supporting components, not by the chip. Test the board in your own application before you rely on it.

Not qualified to any standard, and not for safety-critical roles. Neither the manufacturer nor NZN tests these boards to JEDEC, AEC or any comparable standard, and no ESD, latch-up or life data exists for the assembly. Handle with normal ESD precautions. Do not deploy it in medical, life-support, automotive, aviation or industrial-control applications, or anywhere a failure can cause injury, fire or property damage. Attaching a lithium cell to an undocumented charger raises that bar further: fit protected cells, confirm the terminals yourself, and do not leave a charging board unattended.

5.3 RECOMMENDED OPERATING CONDITIONS

PARAMETER	MIN	TYP	MAX	UNIT
V_{5V} [1]	3.3	5.0	6.0	V
V_{BAT} [2]	3.0	3.7	4.2	V
V_{3V3} [3]	3.0	3.3	3.6	V
Source current capability [4]	500	-	-	mA
3V3 rail draw, external load [5]	-	-	200	mA
Battery charge current [6]	100	-	300	mA
T_A [7]	0	25	55	°C

[1] The 5V pad range is the manufacturer's published figure for this board. [2] One lithium cell on the underside B+ / B- pads. No cells in series, no 5 V here. [3] Whether 3V3 is an output or driven as an input; the chip wants 3.0-3.6 V. [4] The source must hold up under the 355 mA Wi-Fi TX peak of Section 5.5 without droop. [5] Conservative; The manufacturer publishes no regulator rating. [6] 100 mA as shipped, 300 mA with the **BOOST** jumper bridged, which the manufacturer restricts to cells of 500 mAh or larger. [7] Conservative board-level figure, not a qualified rating: the ESP32-S3FH4R2 silicon is rated -40 to +65 °C, that ceiling set by the in-package PSRAM rather than the core, and the regulator, charger, connector and passives are unspecified. See Section 5.2.

5.4 DC ELECTRICAL CHARACTERISTICS

SYM	PARAMETER	MIN	TYP	MAX	UNIT
V_{IH}	High-level input voltage	$0.75\times V_{DD}$	-	$V_{DD}+0.3$	V
V_{IL}	Low-level input voltage	-0.3	-	$0.25\times V_{DD}$	V
V_{OH}	High-level output voltage	$0.8\times V_{DD}$	-	-	V
V_{OL}	Low-level output voltage	-	-	$0.1\times V_{DD}$	V
I_{OH}	Source current, PAD_DRIVER=3	-	40	-	mA
I_{OL}	Sink current, PAD_DRIVER=3	-	28	-	mA
I_{IH}/I_{IL}	Input leakage current	-	-	50	nA
R_{PU}/R_{PD}	Weak pull-up / pull-down	-	45	-	kΩ
C_{IN}	Pin capacitance	-	2	-	pF

V_{DD} = 3.3 V, T_A = 25 °C. V_{OH} and V_{OL} measured into a high-impedance load. Drive strength is software-selectable; PAD_DRIVER=3 is the maximum setting. Total current summed across all pads must stay inside the 1100 mA cumulative limit of Section 5.1.

5.5 CURRENT CONSUMPTION

MODE	CONDITION	TYP	UNIT
Active, TX	802.11b, 1 Mbps, @21 dBm	355	mA
Active, TX	802.11n, HT20, MCS7, @18.5 dBm	282	mA
Active, RX	802.11b/g/n, HT20	95	mA
Active, BLE	TX @0 dBm	80	mA
Modem-sleep	Both cores running at 240 MHz	40	mA
Modem-sleep	Single core at 40 MHz	14	mA
Light-sleep	Wi-Fi off, all GPIO high-Z	240	uA
Deep-sleep	RTC timer plus RTC memory	8	uA
Power off	CHIP_PU held low	1	uA

Chip-level figures at 3.3 V, 25 °C, per the ESP32-S3 series datasheet. **Board-level sleep current is substantially higher**: the regulator and charger quiescent current dominate, and the WS2812 draws roughly 1 mA even showing black because its controller stays powered. Enabling PSRAM also adds refresh current in light sleep. Measure your own board before committing to a battery budget.

5.6 RADIO AND ANALOG

PARAMETER	CONDITION	TYP	UNIT
Wi-Fi centre frequency	Channel range	2412-2484	MHz
Wi-Fi TX power	802.11b, 1 Mbps	21.0	dBm
Wi-Fi TX power	802.11n, HT20, MCS7	18.5	dBm
Wi-Fi RX sensitivity	802.11b, 1 Mbps	-98.5	dBm
Wi-Fi RX sensitivity	802.11g, 54 Mbps	-76.0	dBm
BLE TX power range	Software controlled	-27 to +20	dBm
BLE RX sensitivity	1 Mbps, 30.8% PER	-97	dBm
ADC resolution	2x SAR	12	bit
ADC channels exposed	10 on ADC1, 8 on ADC2	18	ch
ADC DNL / INL	25 °C, radio off	±7 / ±12	LSB
Touch channels exposed	GPIO1-GPIO14	14	ch

Measured at the antenna port with a 3.3 V supply at 25 °C. The onboard ceramic chip antenna adds insertion loss not included above. The ESP32-S3 ADC is **not** linear across the full range; calibrate in software for measurement work. **ADC2 is unavailable while the radio is active**, which rules out GPIO11-GPIO18 for analog whenever Wi-Fi or BLE is on.

6 BOOT CONFIGURATION

BOOT MODE	GPIO0	GPIO46
SPI boot, normal run [1]	1	any
Joint download boot [2]	0	0

STRAPPING PIN	ROLE	STATE AT RESET, AND WHERE IT IS
GPIO0	Boot mode select	Internal pull-up; the BOOT button pulls it low. **No pad** - under the button only
GPIO3	JTAG source select	Floating. Ignored unless the eFuse is burned. **Left rail**
GPIO45	VDD_SPI voltage select	Internal pull-down. **Underside pad**
GPIO46	ROM message printing	Internal pull-down. **Underside pad**

[1] Default path. GPIO0 is held high by its internal pull-up, so the board runs the firmware in flash unless BOOT is pressed at reset. [2] Joint download boot covers USB-Serial-JTAG, USB OTG and UART download. The combination GPIO0 = 0 with GPIO46 = 1 is invalid and gives unpredictable behaviour.

Setup and hold. t_{SU} minimum 0 ms before CHIP_PU rises; t_H minimum 3 ms after CHIP_PU is high, during which the strapping pins are sampled and latched. Once latched they are free for use as ordinary GPIO for the rest of the power cycle.

The GPIO45 trap. This is the one that bites, and on this board it sits on the underside where a carrier PCB is most likely to touch it. GPIO45 sets the VDD_SPI rail that powers the in-package flash and PSRAM. Hold it at the wrong level through reset and the chip powers its memory at the wrong voltage, so it never boots and the fault looks exactly like a dead board. The chip maker normally burns the eFuse that fixes this rail on parts with in-package memory, but do not design around that: **never fit a pull-up, an LED to 3V3, or an idle-high bus signal on GPIO45.**

Design rule. Anything connected to GPIO3, GPIO45 or GPIO46 must not hold the pin at the wrong level through reset. Reflowing this board onto a carrier, or soldering a ribbon to the underside pads, is exactly when that happens. Check the carrier netlist against this table before the first build.

7 APPLICATION AND IMPLEMENTATION

7.1 POWER SUPPLY

USB-C. Normal operation. VBUS feeds the onboard regulator through the 5V net and the charger through its own path.

Battery. Solder one Li-ion or LiPo cell to the **B+** and **B-** pads on the underside, beneath the USB-C connector. They are the only battery connection on the board; the "BAT" mark on the top face labels the charge LED, not a pad. The charger tops the cell up whenever USB-C is connected. Charge current is **100 mA** as shipped and **300 mA** with the **BOOST** jumper bridged, which the manufacturer restricts to cells of 500 mAh or larger. Use protected cells, keep the leads short, and do not leave a charging board unattended.

External 5 V. Apply 3.3-6 V to the 5V pad, per the manufacturer. **Do not back-feed while USB-C is connected to another source** - the manufacturer is explicit that only one of the two may be used at a time.

Direct 3.3 V. Apply a regulated 3.0-3.6 V to the 3V3 pad with 5V, B+ and USB unpowered. This bypasses the regulator and is the lowest-quiescent option for battery work.

Decoupling. Wi-Fi TX draws 355 mA in bursts and the S3 pulls harder than a C3 does, especially with both cores at 240 MHz and PSRAM enabled. Fit at least 100 uF of low-ESR bulk capacitance local to the board on battery or long-lead supplies, or the rail sags and the board brown-out resets mid-transmission.

7.2 ANTENNA AND LAYOUT

The antenna is a ceramic chip part at the end of the board opposite the USB-C connector. Keep that end clear of metal, batteries and ground pour. Overhanging an enclosure edge, or standing the board off a metal chassis by at least 10 mm, materially improves range. A lithium cell taped to the underside of the antenna end is the most common self-inflicted range problem on a board like this.

When you reflow this board onto a carrier, leave the antenna end hanging past the carrier's board edge with no copper underneath it. A ground pour under a chip antenna detunes it and is the usual reason a carrier build has far worse range than the bare board.

Enclosing the board in a metal box will reduce usable range to near zero. Use a plastic enclosure. Keep the antenna end away from USB 3.0 connectors and switching regulators, which are both strong 2.4 GHz noise sources.

The underside pads change the mechanical plan. Twenty of the thirty-five GPIO are on the back face. Seated flat on a breadboard they are covered. If your design needs them, mount the board on headers with clearance underneath, use a socket, or reflow it onto a carrier that routes them out.

7.3 TOOLCHAIN CONFIGURATION

SETTING	VALUE
Arduino IDE board package	**official esp32 board package**
Board selection	**ESP32S3 Dev Module**
USB CDC On Boot	**Enabled**
USB Mode	**Hardware CDC and JTAG**
PSRAM	**QSPI PSRAM**
Flash size	**4 MB (32 Mb)**
Flash mode	**QIO 80 MHz**
Partition scheme	**Default 4 MB with spiffs**
Upload speed	**921600, or 460800 if unstable**
USB driver	**None required**
ESP-IDF target	**esp32s3**
PlatformIO board ID	**esp32-s3-devkitc-1**
ESPHome platform	**esp32, variant esp32s3**
MicroPython port	**ESP32_GENERIC_S3 (SPIRAM)**

No driver install. The ESP32-S3 enumerates as a native USB Serial/JTAG device. Windows 10 (1903 and later), Windows 11, macOS and Linux all supply the CDC driver in the operating system. There is no CH340 or CP2102 bridge on this board.

PSRAM must be switched on. The board package defaults PSRAM to Disabled. Leave it there and the 2 MB is never mapped, which is the single most common complaint about S3 boards of this type.

These menu entries do not appear until an S3 board is selected. USB CDC On Boot, USB Mode, PSRAM and Flash Size are board-specific, so the Tools menu has none of them until Tools > Board is set to ESP32S3 Dev Module. Looking for them first, and concluding the instructions are wrong, is the usual first-run stumble.

7.4 RECOVERY: BOARD DOES NOT ENUMERATE

1. **Use a data-capable USB-C cable.** Charge-only cables are by far the most common cause. Try a second cable before changing anything else.
2. **Check enumeration.** Windows Device Manager should list **USB Serial Device (COMn)** or **USB JTAG/serial debug unit**. On Linux look for **/dev/ttyACM0**; on macOS **/dev/cu.usbmodem***.
3. **Force download mode.** Hold **BOOT**, tap **RST**, release RST, then release BOOT. This asserts GPIO0 low at reset per Section 6. The manufacturer notes the port can drop again between attempts, so repeat the sequence each time.
4. **Check GPIO45 and GPIO46 if the board is on a carrier.** A pull on either underside pad at reset stops the board booting and looks exactly like a hardware failure. Lift the board and retest it bare.
5. **Recover a bricked flash.** In download mode run **esptool erase_flash**, then reflash. This clears firmware that crashes early or reconfigures the USB pins during boot.
6. **Serial monitor shows nothing.** Confirm **USB CDC On Boot** is Enabled. Without it, Serial output is routed to GPIO43 (TX) rather than the USB port.

7.5 ESPTOOL COMMAND REFERENCE

```
# esptool v5 uses hyphenated commands (chip-id, erase-flash).
# On v4 and earlier use the underscore forms (chip_id, erase_flash).
pip install esptool

# Identify the part. Omit --chip so it auto-detects: the chip reports its
# own eFuses, so this cannot be fooled by a listing or an IDE menu.
esptool --port COM5 chip-id

#   Chip type: ESP32-S3 (QFN56) (revision v0.2)
#   Features:  Dual Core + LP Core, 240MHz,
#              Embedded Flash 4MB, Embedded PSRAM 2MB

# Full erase, back to factory blank
esptool --port COM5 erase-flash

# Verify the erase: every byte read back should be 0xFF
esptool --port COM5 read-flash 0x0 0x4000 dump.bin

# Write a firmware image at the base of flash
esptool --port COM5 --baud 460800 write-flash -z 0x0 firmware.bin

# Read the whole 4 MB back to a file
esptool --port COM5 read-flash 0x0 0x400000 backup.bin
```

7.6 DESIGN NOTES AND KNOWN LIMITATIONS

- **Twenty of the thirty-five GPIO are on the underside.** They are covered once the board sits flat on a breadboard. Plan the mechanics before you plan the pinout.
- **PSRAM is off until you enable it.** Set PSRAM to QSPI PSRAM in the Arduino board menu, or `CONFIG_SPIRAM` under ESP-IDF. Without it `ESP.getPsramSize()` returns zero and large allocations fail.
- **The ADC is not linear to the rails.** Usable span depends on the attenuation setting and accuracy degrades at both ends. Calibrate in software for measurement work.
- **There is no independent plain LED.** The WS2812 and the discrete LED share GPIO48, so they cannot be driven separately, and a stock blink sketch on that pin will not behave the way you expect.
- **One ground pad only.** Every return path shares a single GND pin. Keep high-current and fast-switching loops short, and do not use the board as a star point.
- **The charger is only partly documented.** Charge current is published as 100 mA, or 300 mA with BOOST, but there is no cell limit, no protection detail and no charge-termination data. Use protected cells and supervise charging.
- **GPIO45 and GPIO46 are strapping pins on those underside pads.** Nothing you attach may hold either at the wrong level through reset. See Section 6.
- **ADC2 is dead while the radio is on.** That covers GPIO11-GPIO18. Use ADC1 on GPIO1-GPIO10 for any analog reading taken with Wi-Fi or BLE active.
- **Deep-sleep wake is limited to GPIO1-GPIO18 and GPIO21.** Everything from GPIO33 up sits outside the RTC domain and cannot wake the chip.
- **The RGB LED spoils deep sleep.** Its controller draws roughly 1 mA even showing black. Board sleep current will not approach the 8 uA silicon figure with it fitted.
- **USB pins are committed.** GPIO19 and GPIO20 serve native USB and are not exposed, so you cannot free them the way some S3 boards allow. That also means you cannot accidentally lose the console by reconfiguring them.
- **The manufacturer's parameter list describes C3 silicon.** Ignore its "RISC-V single core, 160 MHz, 400 KB SRAM, 11xGPIO, 43 uA" figures, and note it never mentions PSRAM at all: see Section 5.2.

8 MECHANICAL DATA

Board dimensions [1]	**23.5 x 18.0 mm**	Weight	**3.60 g with headers**
Side rail pads	**18, nine per row**	Pad pitch	**2.54 mm**
Edge style	**Castellated, plated through-hole**	Underside GPIO pads	**20, numbered on the silkscreen**
Breadboard compatible	**Yes, one free column per side**	Connector	**USB-C receptacle, edge mounted**
Buttons	**BOOT and RST, tactile**	Battery connection	**B+ / B- solder pads, underside**
Charge jumper	**BOOST, underside, open as shipped**	Indicators	**WS2812 RGB and discrete LED (GPIO48), charge**
Antenna	**Onboard ceramic chip**	Chip package	**QFN56, 7 x 7 mm**
Crystal	**40.000 MHz**	Supplied headers	**2 x 10-way 2.54 mm male, loose [2]**
PCB marking	**"Super Mini / ESP32-S3", underside**		

[1] Dimensions are nominal and given for mechanical fit only. This is not a controlled drawing and no tolerances are implied. Confirm against your own sample before committing an enclosure or PCB footprint to manufacture. [2] The supplied strips are ten-way against nine pads per rail, so snap one position off each or let the surplus overhang.

9 REVISION HISTORY

REV	DATE	CHANGE
1	2026-08-08	Initial release.
2	2026-09-30	Removed supplier and manufacturer names and origin claims from the document text. No specification changed.